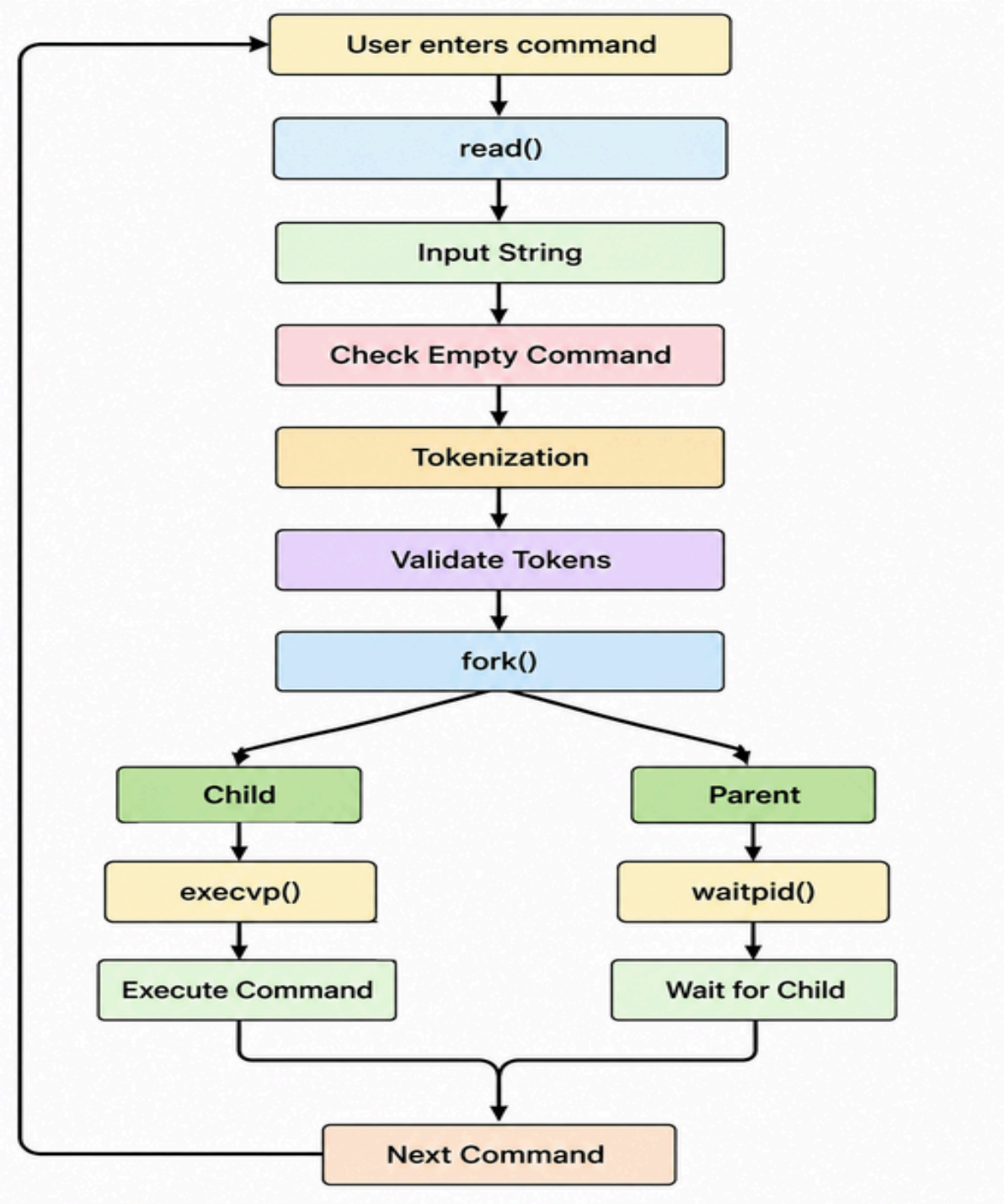
Refer the following link to understand the Parsing and Lexical analysis

https://www.geeksforgeeks.org/compiler-design/introduction-of-lexical-analysis/

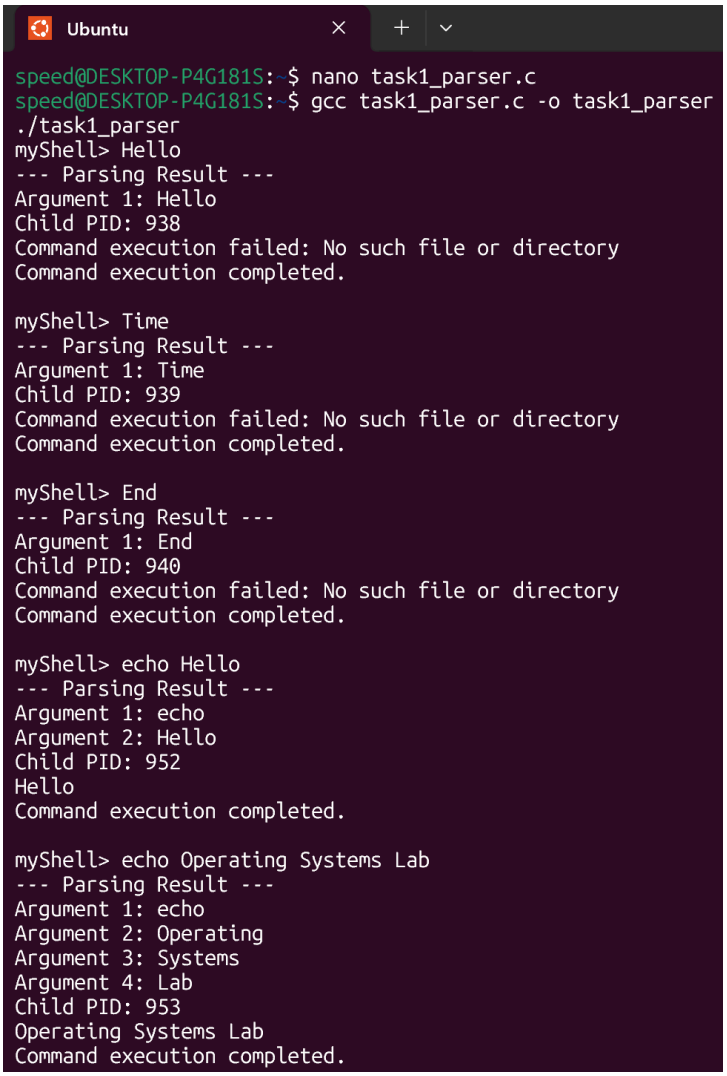
Task 1:

To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output and Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures

Flow Chart:



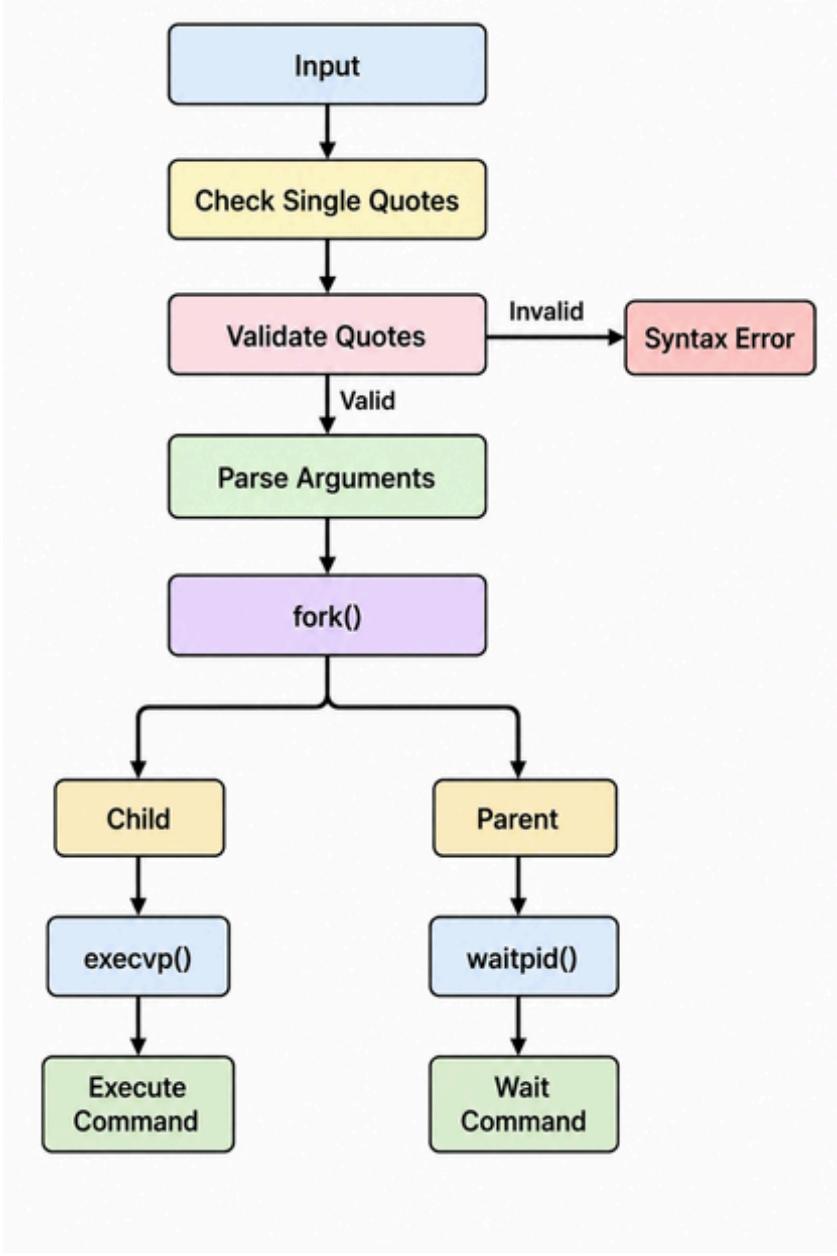
```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

#define BUFFER_SIZE 1024
#define MAX_TOKENS 64

int main() {
    char buffer[BUFFER_SIZE];
    char *tokens[MAX_TOKENS];

    while (1) {
        // 1. Display Prompt
        write(1, "myShell> ", 9);

        // 2. Read user input using read() system call
        memset(buffer, 0, sizeof(buffer));
        int bytes_read = read(0, buffer, sizeof(buffer) - 1);

        if (bytes_read <= 0) {
            printf("\nExiting shell...\n");
            break;
        }

        // Remove trailing newline character
        if (buffer[bytes_read - 1] == '\n') {
            buffer[bytes_read - 1] = '\0';
        }

        // 3. Check for Empty Command
        if (strlen(buffer) == 0) {
            continue;
        }

        // Check for exit
        if (strcmp(buffer, "exit") == 0) {
            printf("Exiting shell.\n");
            break;
        }

        // 4. Tokenization (Splitting input into tokens)
        int token_count = 0;
        char *token = strtok(buffer, " \t\r\n");

        while (token != NULL && token_count < MAX_TOKENS - 1) {
            tokens[token_count++] = token;
            token = strtok(NULL, " \t\r\n");
        }
        tokens[token_count] = NULL; // NULL-terminate the array

        // 5. Validate Tokens & Debug Parsing Output
        if (token_count == 0) {
            continue;
        }

        printf("--- Parsing Result ---\n");
        for (int i = 0; i < token_count; i++) {
            printf("Argument %d: %s\n", i + 1, tokens[i]);
        }

        // 6. Fork and Execute
        pid_t pid = fork();

        if (pid < 0) {
            perror("Fork failed");
        } else if (pid == 0) {
            // Child process
            printf("Child PID: %d\n", getpid());
            if (execvp(tokens[0], tokens) < 0) {
                perror("Command execution failed");
                exit(EXIT_FAILURE);
            }
        } else {
            // Parent process
            int status;
            waitpid(pid, &status, 0);
            printf("Command execution completed.\n\n");
        }
    }

    return 0;
}
```



Task 2:

To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

Flow Chart:



Example Output:

Case1:

myShell> echo Hello

Parsing Result:

Argument 1: echo

Argument 2: Hello

Child PID: 3210

Hello

Command execution completed.

Case2:

myShell> echo 'Hello World

Syntax Error: Unmatched single quote.

```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

// Custom parser to handle single quotes
int parse_single_quotes(char *input, char **args, int *arg_count) {
    int i = 0, j = 0;
    int in_single_quote = 0;
    char current_token[MAX_INPUT];
    int token_len = 0;

    *arg_count = 0;

    while (input[i] != '\0') {
        char c = input[i];

        if (c == '\'') {
            in_single_quote = !in_single_quote; // Toggle quote state
        } else if (c == ' ' && !in_single_quote) {
            // Delimiter reached outside quotes
            if (token_len > 0) {
                current_token[token_len] = '\0';
                args[*arg_count] = strdup(current_token);
                (*arg_count)++;
                token_len = 0;
            }
        } else {
            // Add regular characters to the token
            current_token[token_len++] = c;
        }
        i++;
    }

    // Check for unmatched quote error
    if (in_single_quote) {
        printf("Syntax Error: Unmatched single quote.\n");
        return -1;
    }

    // Save final token if present
    if (token_len > 0) {
        current_token[token_len] = '\0';
        args[*arg_count] = strdup(current_token);
        (*arg_count)++;
    }

    args[*arg_count] = NULL;
    return 0;
}

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];
    int arg_count = 0;

    while (1) {
        printf("myShell> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("\nExiting...\n");
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0) continue;
        if (strcmp(input, "exit") == 0) break;

        // Parse with quote handling
        if (parse_single_quotes(input, args, &arg_count) != 0) {
            continue; // Syntax error occurred
        }

        if (arg_count == 0) continue;

        // Display parsed result
        printf("Parsing Result:\n");
        for (int i = 0; i < arg_count; i++) {
            printf("Argument %d: %s\n", i + 1, args[i]);
        }

        // Fork & Execute
        pid_t pid = fork();
        if (pid < 0) {
            perror("Fork failed");
        } else if (pid == 0) {
            printf("Child PID: %d\n", getpid());
            if (execvp(args[0], args) < 0) {
                perror("Execution error");
                exit(EXIT_FAILURE);
            }
        } else {
            waitpid(pid, NULL, 0);
            printf("Command execution completed.\n\n");
        }

        // Free allocated memory
        for (int i = 0; i < arg_count; i++) {
            free(args[i]);
        }
    }
```

```
    return 0;
}
```

```
Ubuntu
speed@DESKTOP-P4G181S:~$ nano task2_single_quotes.c
speed@DESKTOP-P4G181S:~$ gcc task2_single_quotes.c -o task2_single_quotes
./task2_single_quotes
myShell> echo Operating Systems Lab
Parsing Result:
Argument 1: echo
Argument 2: Operating
Argument 3: Systems
Argument 4: Lab
Child PID: 1077
Operating Systems Lab
Command execution completed.

myShell> exit
speed@DESKTOP-P4G181S:~$
```

Task 3:

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

Example Output:

Case1:

myShell> echo "Hello $USER"

--- Parsed Arguments ---

Argument 1: echo

Argument 2: Hello raghu

Child Process

PID = 3022

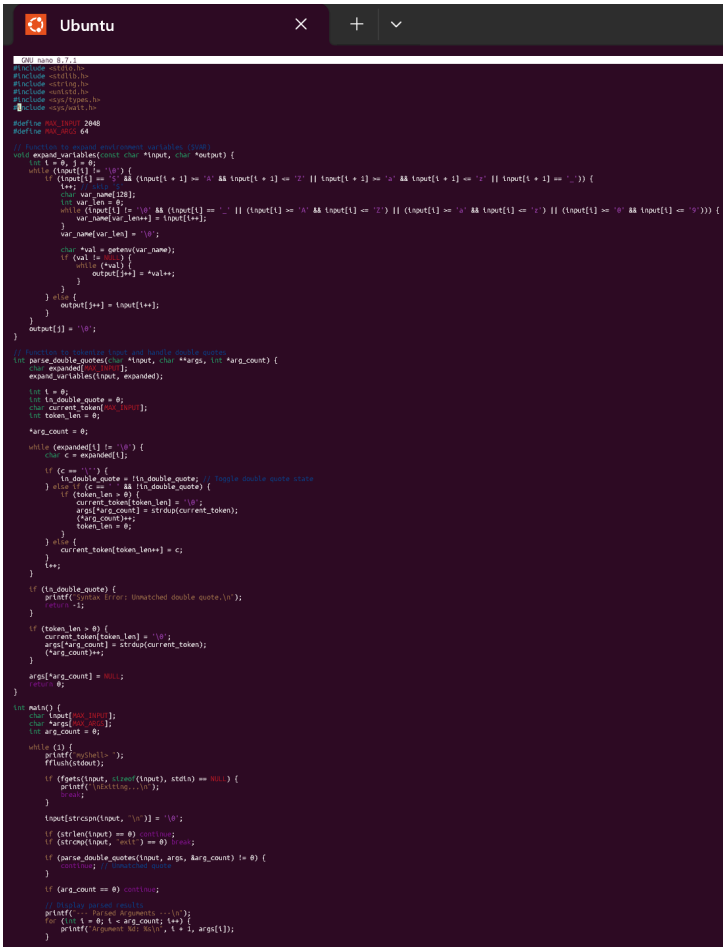
Hello raghu

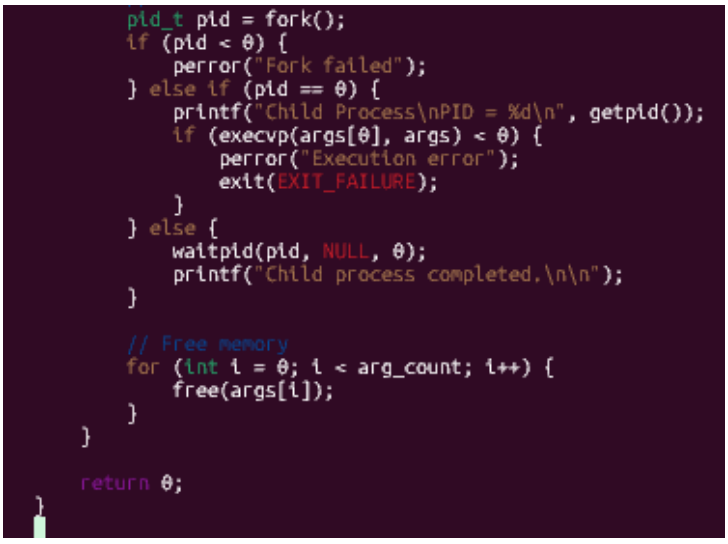
Child process completed.

Case2:

myShell> echo "Hello World

Syntax Error: Unmatched double quote.





```
speed@DESKTOP-P4G181S:~$ nano task3_double_quotes.c
speed@DESKTOP-P4G181S:~$ gcc task3_double_quotes.c -o task3_double_quotes
./task3_double_quotes
myShell> echo Operating Systems Lab
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Operating
Argument 3: Systems
Argument 4: Lab
Child Process
PID = 1121
Operating Systems Lab
Child process completed.
```

```
myShell> echo 'Hello World'
--- Parsed Arguments ---
Argument 1: echo
Argument 2: 'Hello
Argument 3: World'
Child Process
PID = 1183
'Hello World'
Child process completed.

myShell>
```